The empirical recurrence for OEIS A264516: recovering a conjecture that is not written down

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Abstract

OEIS A264516 records "Empirical recurrence of order 84", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 1024$ vertices and satisfies some monic recurrence of order at most S; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 84, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 84 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A264516 is "Number of (n+1) X (4+1) arrays of permutations of 0..n*5+4 with each element having directed index change -1,0 0,2 -1,-2 or 1,0.". It has offset 1 and begins

$$2, 2, 42, 228, 1644, 10368, 65182, 414414, \ \dots$$

The entry states:

> Empirical recurrence of order 84 (see link above)

As of the "Last modified" line on the live entry (Apr 21 2021 13:45:52, revision 6) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1024$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S =$ 1024.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 2048$ exact terms of the model returns order 84 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{84} c_i\, a(n-i),$$

c_1	4	c_{22}	1182301310	c_{43}	−1268853264	c_{64}	−453210430
c_2	46	c_{23}	980431632	c_{44}	11553685878	c_{65}	103599474
c_3	−162	c_{24}	−2383944670	c_{45}	3480567568	c_{66}	131640593
c_4	−769	c_{25}	−1920813272	c_{46}	−8827860683	c_{67}	−12089012
c_5	3246	c_{26}	3804117496	c_{47}	−4895850428	c_{68}	−29708092
c_6	6983	c_{27}	3053179766	c_{48}	4897162294	c_{69}	−4361588
c_7	−36718	c_{28}	−4814349472	c_{49}	5463063972	c_{70}	5642561
c_8	−40546	c_{29}	−4130018726	c_{50}	−692722037	c_{71}	3144158
c_9	249934	c_{30}	4624993035	c_{51}	−5418119852	c_{72}	−1002475
c_{10}	192846	c_{31}	4950624320	c_{52}	−2737843569	c_{73}	−1093946
c_{11}	−1093946	c_{32}	−2737843569	c_{53}	4950624320	c_{74}	192846
c_{12}	−1002475	c_{33}	−5418119852	c_{54}	4624993035	c_{75}	249934
c_{13}	3144158	c_{34}	−692722037	c_{55}	−4130018726	c_{76}	−40546
c_{14}	5642561	c_{35}	5463063972	c_{56}	−4814349472	c_{77}	−36718
c_{15}	−4361588	c_{36}	4897162294	c_{57}	3053179766	c_{78}	6983
c_{16}	−29708092	c_{37}	−4895850428	c_{58}	3804117496	c_{79}	3246
c_{17}	−12089012	c_{38}	−8827860683	c_{59}	−1920813272	c_{80}	−769
c_{18}	131640593	c_{39}	3480567568	c_{60}	−2383944670	c_{81}	−162
c_{19}	103599474	c_{40}	11553685878	c_{61}	980431632	c_{82}	46
c_{20}	−453210430	c_{41}	−1268853264	c_{62}	1182301310	c_{83}	4
c_{21}	−383548266	c_{42}	−12517761315	c_{63}	−383548266	c_{84}	−1

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 84, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $84 + 4$ terms removed returns order 84 as well, so $\deg q_{\min} = 84 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 21 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $84+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 84 that this sequence satisfies — and that family turns out to have exactly one member.

References

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